

we work with the gradient, which is the vector of partial derivatives. The i th component of the vector gradient has the form

$$\frac{\partial l(W)}{\partial w_i} = \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1 | X^l, W))$$

where $\hat{P}(Y^l | X^l, W)$ is the Logistic Regression prediction using equations (24) and (25) and the weights W . To accommodate weight w_0 , we assume an imaginary $X_0 = 1$ for all l . This expression for the derivative has an intuitive interpretation: the term inside the parentheses is simply the prediction error; that is, the difference between the observed Y^l and its predicted probability! Note if $Y^l = 1$ then we wish for $\hat{P}(Y^l = 1 | X^l, W)$ to be 1, whereas if $Y^l = 0$ then we prefer that $\hat{P}(Y^l = 1 | X^l, W)$ be 0 (which makes $\hat{P}(Y^l = 0 | X^l, W)$ equal to 1). This error term is multiplied by the value of X_i^l , which accounts for the magnitude of the $w_i X_i^l$ term in making this prediction.

Given this formula for the derivative of each w_i , we can use standard gradient ascent to optimize the weights W . Beginning with initial weights of zero, we repeatedly update the weights in the direction of the gradient, on each iteration changing every weight w_i according to

$$w_i \leftarrow w_i + \eta \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1 | X^l, W))$$

where η is a small constant (e.g., 0.01) which determines the step size. Because the conditional log likelihood $l(W)$ is a concave function in W , this gradient ascent procedure will converge to a global maximum. Gradient ascent is described in greater detail, for example, in Chapter 4 of Mitchell (1997). In many cases where computational efficiency is important it is common to use a variant of gradient ascent called conjugate gradient ascent, which often converges more quickly.

3.3 Regularization in Logistic Regression

Overfitting the training data is a problem that can arise in Logistic Regression, especially when data is very high dimensional and training data is sparse. One approach to reducing overfitting is *regularization*, in which we create a modified “penalized log likelihood function,” which penalizes large values of W . One approach is to use the penalized log likelihood function

$$W \leftarrow \arg \max_W \sum_l \ln P(Y^l | X^l, W) - \frac{\lambda}{2} \|W\|^2$$

which adds a penalty proportional to the squared magnitude of W . Here λ is a constant that determines the strength of this penalty term.

Modifying our objective by adding in this penalty term gives us a new objective to maximize. It is easy to show that maximizing it corresponds to calculating the MAP estimate for W under the assumption that the prior distribution $P(W)$ is

a Normal distribution with mean zero, and a variance related to $1/\lambda$. Notice that in general, the MAP estimate for W involves optimizing the objective

$$\sum_l \ln P(Y^l | X^l, W) + \ln P(W)$$

and if $P(W)$ is a zero mean Gaussian distribution, then $\ln P(W)$ yields a term proportional to $\|W\|^2$.

Given this penalized log likelihood function, it is easy to rederive the gradient descent rule. The derivative of this penalized log likelihood function is similar to our earlier derivative, with one additional penalty term

$$\frac{\partial l(W)}{\partial w_i} = \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1 | X^l, W)) - \lambda w_i$$

which gives us the modified gradient descent rule

$$w_i \leftarrow w_i + \eta \sum_l X_i^l (Y^l - \hat{P}(Y^l = 1 | X^l, W)) - \eta \lambda w_i \quad (26)$$

In cases where we have prior knowledge about likely values for specific w_i , it is possible to derive a similar penalty term by using a Normal prior on W with a non-zero mean.

3.4 Logistic Regression for Functions with Many Discrete Values

Above we considered using Logistic Regression to learn $P(Y|X)$ only for the case where Y is a boolean variable. More generally, if Y can take on any of the discrete values $\{y_1, \dots, y_K\}$, then the form of $P(Y = y_k | X)$ for $Y = y_1, Y = y_2, \dots, Y = y_{K-1}$ is:

$$P(Y = y_k | X) = \frac{\exp(w_{k0} + \sum_{i=1}^n w_{ki} X_i)}{1 + \sum_{j=1}^{K-1} \exp(w_{j0} + \sum_{i=1}^n w_{ji} X_i)} \quad (27)$$

When $Y = y_K$, it is

$$P(Y = y_K | X) = \frac{1}{1 + \sum_{j=1}^{K-1} \exp(w_{j0} + \sum_{i=1}^n w_{ji} X_i)} \quad (28)$$

Here w_{ji} denotes the weight associated with the j th class $Y = y_j$ and with input X_i . It is easy to see that our earlier expressions for the case where Y is boolean (equations (16) and (17)) are a special case of the above expressions. Note also that the form of the expression for $P(Y = y_K | X)$ assures that $[\sum_{k=1}^K P(Y = y_k | X)] = 1$.

The primary difference between these expressions and those for boolean Y is that when Y takes on K possible values, we construct $K - 1$ different linear expressions to capture the distributions for the different values of Y . The distribution for

the final, K th, value of Y is simply one minus the probabilities of the first $K - 1$ values.

In this case, the gradient descent rule with regularization becomes:

$$w_{ji} \leftarrow w_{ji} + \eta \sum_l X_i^l (\delta(Y^l = y_j) - \hat{P}(Y^l = y_j | X^l, W)) - \eta \lambda w_{ji} \quad (29)$$

where $\delta(Y^l = y_j) = 1$ if the l th training value, Y^l , is equal to y_j , and $\delta(Y^l = y_j) = 0$ otherwise. Note our earlier learning rule, equation (26), is a special case of this new learning rule, when $K = 2$. As in the case for $K = 2$, the quantity inside the parentheses can be viewed as an error term which goes to zero if the estimated conditional probability $\hat{P}(Y^l = y_j | X^l, W)$ perfectly matches the observed value of Y^l .

4 Relationship Between Naive Bayes Classifiers and Logistic Regression

To summarize, Logistic Regression directly estimates the parameters of $P(Y|X)$, whereas Naive Bayes directly estimates parameters for $P(Y)$ and $P(X|Y)$. We often call the former a discriminative classifier, and the latter a generative classifier.

We showed above that the assumptions of one variant of a Gaussian Naive Bayes classifier imply the parametric form of $P(Y|X)$ used in Logistic Regression. Furthermore, we showed that the parameters w_i in Logistic Regression can be expressed in terms of the Gaussian Naive Bayes parameters. In fact, if the GNB assumptions hold, then asymptotically (as the number of training examples grows toward infinity) the GNB and Logistic Regression converge toward identical classifiers.

The two algorithms also differ in interesting ways:

- When the GNB modeling assumptions do not hold, Logistic Regression and GNB typically learn different classifier functions. In this case, the asymptotic (as the number of training examples approach infinity) classification accuracy for Logistic Regression is often better than the asymptotic accuracy of GNB. Although Logistic Regression is consistent with the Naive Bayes assumption that the input features X_i are conditionally independent given Y , it is not rigidly tied to this assumption as is Naive Bayes. Given data that disobeys this assumption, the conditional likelihood maximization algorithm for Logistic Regression will adjust its parameters to maximize the fit to (the conditional likelihood of) the data, even if the resulting parameters are inconsistent with the Naive Bayes parameter estimates.
- GNB and Logistic Regression converge toward their asymptotic accuracies at different rates. As Ng & Jordan (2002) show, GNB parameter estimates converge toward their asymptotic values in order $\log n$ examples, where n is the dimension of X . In contrast, Logistic Regression parameter estimates

converge more slowly, requiring order n examples. The authors also show that in several data sets Logistic Regression outperforms GNB when many training examples are available, but GNB outperforms Logistic Regression when training data is scarce.

5 What You Should Know

The main points of this chapter include:

- We can use Bayes rule as the basis for designing learning algorithms (function approximators), as follows: Given that we wish to learn some target function $f : X \rightarrow Y$, or equivalently, $P(Y|X)$, we use the training data to learn estimates of $P(X|Y)$ and $P(Y)$. New X examples can then be classified using these estimated probability distributions, plus Bayes rule. This type of classifier is called a *generative* classifier, because we can view the distribution $P(X|Y)$ as describing how to generate random instances X conditioned on the target attribute Y .
- Learning Bayes classifiers typically requires an unrealistic number of training examples (i.e., more than $|X|$ training examples where X is the instance space) unless some form of prior assumption is made about the form of $P(X|Y)$. The *Naive Bayes* classifier assumes all attributes describing X are conditionally independent given Y . This assumption dramatically reduces the number of parameters that must be estimated to learn the classifier. Naive Bayes is a widely used learning algorithm, for both discrete and continuous X .
- When X is a vector of discrete-valued attributes, Naive Bayes learning algorithms can be viewed as linear classifiers; that is, every such Naive Bayes classifier corresponds to a hyperplane decision surface in X . The same statement holds for Gaussian Naive Bayes classifiers if the variance of each feature i is assumed to be independent of the class k (i.e., if $\sigma_{ik} = \sigma_i$).
- Logistic Regression is a function approximation algorithm that uses training data to directly estimate $P(Y|X)$, in contrast to Naive Bayes. In this sense, Logistic Regression is often referred to as a *discriminative* classifier because we can view the distribution $P(Y|X)$ as directly discriminating the value of the target value Y for any given instance X .
- Logistic Regression is a linear classifier over X . The linear classifiers produced by Logistic Regression and Gaussian Naive Bayes are identical in the limit as the number of training examples approaches infinity, *provided* the Naive Bayes assumptions hold. However, if these assumptions do not hold, the Naive Bayes bias will cause it to perform less accurately than Logistic Regression, in the limit. Put another way, Naive Bayes is a learning algorithm with greater bias, but lower variance, than Logistic Regression. If

this bias is appropriate given the actual data, Naive Bayes will be preferred. Otherwise, Logistic Regression will be preferred.

- We can view function approximation learning algorithms as statistical estimators of functions, or of conditional distributions $P(Y|X)$. They estimate $P(Y|X)$ from a sample of training data. As with other statistical estimators, it can be useful to characterize learning algorithms by their bias and expected variance, taken over different samples of training data.

6 Further Reading

Wasserman (2004) describes a Reweighted Least Squares method for Logistic Regression. Ng and Jordan (2002) provide a theoretical and experimental comparison of the Naive Bayes classifier and Logistic Regression.

EXERCISES

1. At the beginning of the chapter we remarked that “A hundred training examples will usually suffice to obtain an estimate of $P(Y)$ that is within a few percent of the correct value.” Describe conditions under which the 95% confidence interval for our estimate of $P(Y)$ will be ± 0.02 .
2. Consider learning a function $X \rightarrow Y$ where Y is boolean, where $X = \langle X_1, X_2 \rangle$, and where X_1 is a boolean variable and X_2 a continuous variable. State the parameters that must be estimated to define a Naive Bayes classifier in this case. Give the formula for computing $P(Y|X)$, in terms of these parameters and the feature values X_1 and X_2 .
3. In section 3 we showed that when Y is Boolean and $X = \langle X_1 \dots X_n \rangle$ is a vector of continuous variables, then the assumptions of the Gaussian Naive Bayes classifier imply that $P(Y|X)$ is given by the logistic function with appropriate parameters W . In particular:

$$P(Y = 1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)}$$

and

$$P(Y = 0|X) = \frac{\exp(w_0 + \sum_{i=1}^n w_i X_i)}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)}$$

Consider instead the case where Y is Boolean and $X = \langle X_1 \dots X_n \rangle$ is a vector of *Boolean* variables. Prove for this case also that $P(Y|X)$ follows this same form (and hence that Logistic Regression is also the discriminative counterpart to a Naive Bayes generative classifier over Boolean features).

Hints:

- Simple notation will help. Since the X_i are Boolean variables, you need only one parameter to define $P(X_i|Y = y_k)$. Define $\theta_{i1} \equiv P(X_i = 1|Y = 1)$, in which case $P(X_i = 0|Y = 1) = (1 - \theta_{i1})$. Similarly, use θ_{i0} to denote $P(X_i = 1|Y = 0)$.
- Notice with the above notation you can represent $P(X_i|Y = 1)$ as follows

$$P(X_i|Y = 1) = \theta_{i1}^{X_i} (1 - \theta_{i1})^{(1-X_i)}$$

Note when $X_i = 1$ the second term is equal to 1 because its exponent is zero. Similarly, when $X_i = 0$ the first term is equal to 1 because its exponent is zero.

4. (based on a suggestion from Sandra Zilles). This question asks you to consider the relationship between the MAP hypothesis and the Bayes optimal hypothesis. Consider a hypothesis space H defined over the set of instances X , and containing just two hypotheses, $h1$ and $h2$ with equal prior probabilities $P(h1) = P(h2) = 0.5$. Suppose we are given an arbitrary set of training data D which we use to calculate the posterior probabilities $P(h1|D)$ and $P(h2|D)$. Based on this we choose the MAP hypothesis, and calculate the Bayes optimal hypothesis. Suppose we find that the Bayes optimal classifier is not equal to either $h1$ or to $h2$, which is generally the case because the Bayes optimal hypothesis corresponds to “averaging over” all hypotheses in H . Now we create a new hypothesis $h3$ which is equal to the Bayes optimal classifier with respect to H , X and D ; that is, $h3$ classifies each instance in X exactly the same as the Bayes optimal classifier for H and D . We now create a new hypothesis space $H' = \{h1, h2, h3\}$. If we train using the same training data, D , will the MAP hypothesis from H' be $h3$? Will the Bayes optimal classifier with respect to H' be equivalent to $h3$? (Hint: the answer depends on the priors we assign to the hypotheses in H' . Can you give constraints on these priors that assure the answers will be yes or no?)

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